GLOWUP BEAUTY BOOKING PLATFORM

SOFTWARE REQUIREMENT SPECIFICATION & TECHNICAL ARCHITECTURE BLUEPRINT

Lead Systems Architect & Administrator:

JOAN ATIMAKU

Department of Software Engineering & Mobile Information Systems

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1. Problem Statement

The beauty and personal care service sector in developing regions continues to face severe operational bottlenecks due to heavy reliance on manual, paper-based scheduling and unorganized appointment tracking. Clients seeking quality beauty treatments—such as hair styling, barbering, nail care, skincare, makeup, lash extensions, and spa treatments—experience widespread friction. Common challenges include lack of price transparency, long physical queue wait times, inability to search local specialists by category or location, and absence of reliable appointment reminder mechanisms.

Concurrently, salon business owners and independent beauty technicians suffer from scheduling overlap conflicts, unrecorded walk-in sales, inability to track staff performance, and lack of automated commission reporting tools. Existing web-based booking solutions fail to operate in areas with intermittent internet connectivity, leading to data loss and uncoordinated bookings. Therefore, there is an urgent need for an offline-first, multi-role mobile application that unifies clients, salon owners, and independent specialists onto a single, reliable digital ecosystem.

2. Project Objectives

2.1 Main Objective

To design, develop, and deploy an offline-first, multi-role native Android application—the **GlowUp Beauty Platform**—that delivers an intuitive, conflict-free appointment booking journey for clients while providing business owners and independent specialists with production-grade management and reporting tools.

2.2 Specific Objectives

1. To construct a thread-safe, offline-first SQLite relational database layer with foreign key constraints, password hashing, and mathematical time-slot conflict detection algorithms.

2. To implement a unified single application architecture capable of dynamically routing users to four distinct role-driven interfaces (Customer/Client, Salon Owner, Technician/Independent Specialist, and System Administrator).
3. To incorporate localized Ugandan Shillings (UGX) pricing (standardized under 200,000 UGX), real-time query search, and 7 core beauty category vector icon assets.
4. To build an interactive referral program engine providing 5% booking discounts (`GLOWUP5`) and 500 UGX reward credits via direct WhatsApp, SMS, Email, and Android System Share Sheet integrations.
5. To design a dedicated System Admin Dashboard for **Joan Atimaku** providing platform revenue analytics, global catalogue controls, and ticket management (Pending, Resolved/Worked Upon, Declined).

3. Significance and Justification

The implementation of the GlowUp Beauty Platform carries substantial technical, economic, and operational significance:

- **Economic Empowerment for Independent Technicians**: By allowing independent service providers to register directly (`RegisterActivity`) and manage their own profiles, schedules, and 70% direct booking commissions, the platform fosters micro-entrepreneurship and income transparency.
- **Operational Efficiency for Salon Owners**: Salon managers gain real-time visibility into total platform volume, walk-in sales tracking, staff scheduling, and automated revenue/commission distribution reports.
- **Seamless Client Experience**: Clients benefit from instant pre-filled live date/time selection, native `DatePickerDialog` and `TimePickerDialog` interfaces, zero time-slot double bookings, and transparent UGX pricing.
- **Offline-First Reliability**: By utilizing thread-safe SQLite persistence (`AppDatabaseHelper.kt`), the application guarantees full operational capability regardless of network fluctuations.

4. Functional Requirements

Requirement ID	Functional Module	Detailed Specification
FR-1	User Authentication & Roles	Supports registration and login with password show/hide toggle (`btnTogglePassword`), system autofill hints, and 4 role dashboards (`CLIENT`, `OWNER`, `TECHNICIAN`, `ADMIN`).
FR-2	Admin Interface (Joan Atimaku)	Login with `JOAN ATIMAKU` / `123456` routes to Client view first, featuring an Admin header tab that launches `AdminDashboardActivity` for system analytics and global control.
FR-3	Real-time Search & Filtering	`etSearchServices` enables live text filtering by name, category, or description across all 7 beauty categories (Hair, Barber, Nails, Skin Care, Makeup, Lash, Spas).
FR-4	Category State Binding	Tapping any category button dynamically highlights it with the primary plum background (`#6D2348`) and white text, instantly resetting unselected buttons.
FR-5	Specialist Selection	`SpecialistSelectionActivity` allows clients to pick top-rated specialists (Musiimenta Zam, Poni Ritah Minga, Omoding Julius) with rating badges and distance metrics.
FR-6	Conflict-Free Booking	Automatically pre-fills live current date (YYYY-MM-DD) and time (HH:mm), attaches native date/time pickers, and enforces mathematical time-slot overlap prevention.
FR-7	Ticket Status Management	`OrderDetailsActivity` enables Admins and Staff to update order/booking tickets between `Pending`, `Resolved / Worked Upon`, and `Declined`.
FR-8	Customer Profile & Preferences	`CustomerProfileActivity` handles personal details, saved addresses, saved salon wishlist, referral rewards, and customer support ticket submissions.
FR-9	Referral & Multi-Sharing	Tapping Special Offer opens a Bottom Sheet dialog offering 5% discounts (`GLOWUP5`), 500 UGX credit, 1-tap copy, WhatsApp, SMS, Email, and System Share Sheet sharing.

Requirement ID	Functional Module	Detailed Specification
FR-10	Business Owner Dashboard	`OwnerDashboardActivity` tracks total revenue in UGX, active booking counts, 70% staff commissions, catalogue pricing, and walk-in sales.
FR-11	Technician Availability Toggle	`TechnicianDashboardActivity` allows specialists to toggle availability status live between "Available for Bookings" and "On Break / Unavailable".
FR-12	Reviews & Ratings	Only verified customers with completed appointments can submit 1–5 star ratings and review comments (`ReviewActivity`).

5. Non-Functional Requirements

Requirement ID	Category	Technical Specification
NFR-1	Performance	All UI activities must load and render in under 1.5 seconds. Query execution must perform smoothly on mid-range mobile devices.
NFR-2	Offline Reliability	Full CRUD functionality operates locally via SQLite without external network dependency. Database warm-up occurs safely in `GlowUpApplication`.
NFR-3	Security & Hashing	Passwords are stored as SHA-256 cryptographic hashes (`PasswordHasher.kt`). Database handles case-insensitive username matching (`COLLATE NOCASE`).
NFR-4	Usability & Ergonomics	All interactive controls observe a minimum 48dp touch target. Header tabs use horizontal scroll views to prevent button clipping.
NFR-5	Data Integrity	Foreign key constraints and CASCADE deletes are enforced (`db.setForeignKeyConstraintsEnabled(true)`). Multi-step writes use transactions.
NFR-6	Maintainability	Employs a Singleton database pattern (`getInstance`), repositories, data models, and clean XML View layouts.
NFR-7	Compatibility	Tested and fully compatible with Android OS API level 26 (Android 8.0 Oreo) and above across various screen densities.

6. System Architecture & Database Schema

The database layer operates under version 6 of `glowup.db`. The primary relational tables and test accounts are structured as follows:

Pre-Seeded Account Roster

Role	Full Name	Username	Password
ADMIN	Joan Atimaku	JOAN ATIMAKU	123456
OWNER	Omoding Dickens Derrick	owner1	password123
TECHNICIAN	Musiimenta Zam	tech1	password123
INDEPENDENT TECH	Omoding Julius	indie1	password123
EMPLOYEE	Poni Ritah Minga	employee1	password123
CLIENT	Tamale Matia	client1	password123

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